

0506 최종 제출본

파일: [JUNG Song_yi_Developing_an_AI_Cultural_Heritage_Docent_Web_P.dhc](#)
텍스트 원본 그대로 복붙

The main title of your contribution – **this field cannot be edited**

Developing an AI Cultural Heritage Docent Web Prototype: Internalizing Institutional Guidelines for Responsible Curation

Add a subtitle (optional):

Song-yi Jung (songi8812@gmail.com), The Academy of Korean Studies, Republic of Korea

Iro Lim (bkksg.studio@gmail.com), The Academy of Korean Studies, Republic of Korea

Baro Kim (ddokbaro@gmail.com), The Academy of Korean Studies, Republic of Korea

Author information – **this field cannot be edited**

Main body – enter your text and images here:

Background and Objectives

This study proposes a Graph-RAG-based AI Smart Docent web prototype that integrates a semantic knowledge graph with generative AI to overcome both LLM hallucination and the structural constraints of physical signage. Current cultural heritage information services face two structural limitations. First, traditional physical interpretive media remain confined to unilateral information delivery, providing uniform content despite the diversity of visitors' ages, prior knowledge, and languages. Second, general-purpose LLMs introduced as an alternative tend to generate factually inaccurate information due to their lack of domain-specific knowledge, thereby undermining the epistemic reliability required in cultural heritage interpretation.

Selection of the Empirical Subject

This study selects the Hanging Painting of Janggoksa Temple (Maitreya Buddha, 1673, National Treasure) as a single empirical case. The work depicts dozens of iconographic figures—buddhas, bodhisattvas, and arhats—on a single canvas, and its *hwagi* (畫記) explicitly records diverse donation categories (materials, pigments, ritual offerings, production stages) and a roster of participating monk-painters. This data composition, in which iconographic classification and person-relation information richly coexist, provides suitable conditions for validating the proposed integration of a

thesaurus and a relation-expressive knowledge graph (Jeong 2012; Park 2012; Korea Heritage Service 2022). The methodology is first validated through this work and then progressively extended to Buddhist sculpture, architecture, and archaeological sites, establishing a foundation for a generalizable generative docent model across Korean cultural heritage.

Methodological Orientation

This study formalizes humanities data—basic specifications, hwagi records, personal records, and thesauri—into RDF/OWL-based semantic data as a representational layer for scholarly knowledge sharing, and converts these data into a graph database for Graph-RAG inference. A generative AI Smart Docent model is then developed using verified data retrieved through natural language queries, and implemented as a web-based prototype.

Related Work and Differentiation

This study builds on prior research that leverages digital technologies to expand museums beyond their physical boundaries, including Semantic Web-based meta-archives (Jung 2023), immersive digital exhibitions in ubiquitous environments (Yoo 2014), and personalized smart docent services (Gu / Shin 2016). While generative AI has recently enabled highly personalized information delivery without pre-defined scenarios, its hallucination problem remains a critical limitation in domains requiring historical accuracy (Natale et al. 2025). Therefore, recent studies (MuseRAG++, Cultural Heritage Assistant) emphasize grounding responses in expert-verified data to ensure responsible curation and reliability (Trichopoulos et al. 2023; Nazar / Acosta 2023; Wang et al. 2025; Hu 2025).

Data Engineering and Knowledge Structuring

The core of this study lies in restructuring analog and unstructured source resources into machine-readable forms inferable by generative AI, thereby technically mitigating AI hallucination. (1) The signage guidelines issued by the Korea Heritage Service are formalized as rule-based constraints embedded into the system prompt, specifying a priori the narrative principles the AI must observe (Korea Heritage Service 2024). (2) Four categories of heterogeneous data—basic specifications, hwagi records, personal records, and thesauri—are formalized as RDF/OWL ontologies, leveraging SKOS for the iconographic thesaurus and CIDOC-CRM for semantic relations among persons, works, places, and events. The formalized data are then converted and loaded into a property graph in Neo4j for Graph-RAG inference, with iterative refinement: load results are validated via Cypher and the RDF/OWL source is revised on discrepancies. This establishes a dual structure: RDF/OWL functions as the representational standard for knowledge modeling and scholarly sharing, while the Neo4j property graph supports real-time inference and query response. (3) Multilingual designations and alternative names (*imyeong*, 異名) are collected and mapped from authoritative sources including the Digital Dictionary of Buddhism

(Muller 2009), the Encyclopedia of Korean Culture, the Korean Heritage Portal, and the hwagi of the Janggoksa Hanging Painting of Maitreya Buddha (Korea Heritage Service 2022). Each value receives a source-identifying prefix to secure scholarly accuracy and provenance, technically supporting the epistemic reliability of AI-generated interpretation.

System Implementation and Inference Pipeline

■ System_architecture_0503.jpg

Figure 1. System Architecture of the Interactive AI Cultural Heritage Docent

The Graph-RAG of this study is a RAG variant that uses structured query results from the graph database as grounding sources, aligning LLM responses with verified facts (**see Figure 1**). The backend server classifies natural language queries by type and selectively executes pre-defined Cypher queries, reliably retrieving core humanistic elements such as participating artists, dates, iconography, and hwagi records. In the generation stage, the LangChain framework integrates retrieved graph data into the prompt, with Google Gemini as the generative model. The system prompt incorporates the Korea Heritage Service guidelines, and the system adjusts narrative style and difficulty according to user expertise (children, general audiences, domain experts) and language preference, forming a customized generation mechanism in which the same grounding data can be reconfigured for different contexts.

Interaction and User Experience

Implemented with a Next.js front-end and a FastAPI back-end, the prototype offers a button-based interface for question selection and structured, data-grounded interpretation. The study further proposes an AI Smart Docent function in which core iconographic elements serve as SVG-based clickable images; selecting an element triggers real-time retrieval of associated graph data and generates a customized interpretation. This multimodal approach—integrating visual, textual, and structured data—lays a foundation for future extension to AR and 3D-based exhibition environments responsive to visitor interaction.

Conclusion and Expected Contributions

This study recontextualizes professional docent interpretation as a visitor-driven interactive experience. By formalizing dispersed humanities knowledge into a dual structure of RDF/OWL-based representation and graph-database-based inference with integrated real-time access, the system simultaneously supports scholarly knowledge sharing and AI-driven inference. By using verified semantic data as a grounding source to mitigate AI hallucination, the study presents the technical feasibility of responsible curation in digital environments. Ultimately, the model contributes a generalizable framework extensible to diverse cultural heritage domains, advancing the practical use of digital heritage and scholarly methodologies.

The system's effectiveness, however, requires further validation through user evaluation and quantitative testing; the constructed RDF/OWL ontology and implementation code will be released with a forthcoming journal article to support reproducibility and collaborative extension within the digital cultural heritage research community.

References

Enter your references here:

Books

Kim, Jung-hee (2019): Hanguk Bulgyo Misulsa [한국불교미술사, History of Korean Buddhist Art]. Sechang Publishing Company.

Kim, Won-ryong / Ahn, Hwi-joon (2003): Hanguk Misulsa [한국미술사, History of Korean Art]. Sigongsa.

Korea Heritage Service (2019): Guidelines for Writing English Cultural Heritage Signage (Rev. ed.).

Korea Heritage Service (2022): Daehyeong bulhwa jeongmil josa bogoseo 51: Gukbo Janggoksa Mireukbul gwaebultaeng [대형불화 정밀조사 보고서 51 _ 국보 장곡사 미륵불괘불탱, Research report of large Buddhist painting gwaebultaeng 51: Hanging Painting of Janggoksa Temple (Maitreya Buddha)]. Korea Heritage Service.

Korea Heritage Service (2024): Gukga yusan annaepan tonghap garideuraen [국가유산 안내판 통합 가이드라인]. Korea Heritage Service.

Moon, Myung-dae (2021): Hanguk Bulgyo Hoehwasa [한국불교회화사, History of Korean Buddhist painting]. Youlhwadang.

Articles & Proceedings

Bonetti, Ana / Salcedo-Puche, Adrian / Vila-Francés, Joan / Benavent-Garcia, Xavier / Fernández-Vargas, Elena / Magdalena-Benedito, Ramon / Soria-Olivas, Emilio (2025): “An Agent-Based RAG Architecture for Intelligent Tourism Assistance: The Valencia Case Study”, in: Preprints. DOI: 10.20944/preprints202511.0576.v1

Gu, Ja-yoon / Shin, Dong-heui (2016): “The Importance of Robot Personality in a Museum Context” [박물관 맥락에서의 로봇성격 설정의 중요성 연구], in: Journal of the Korea Contents Association [한국콘텐츠학회논문지], 16(3), 184–197. DOI: 10.5392/JKCA.2016.16.03.184

Hu, Albert (2025): “MuseRAG++: A Deep Retrieval-Augmented Generation Framework for Semantic Interaction and Multi-Modal Reasoning in Virtual Museums”, in: Research Square. DOI: 10.21203/rs.3.rs-7281889/v1

Jeong, Myoung-hee (2012): "Iconographic overlap in the gwaebultaeng (掛佛幀, hanging Buddhist painting) at Janggoksa Temple, and its significance" [長谷寺 靈山大會掛佛幀>에 보이는 도상의 중첩과 그 의미], in: Bulgyo Misulsahak [불교미술사학, Journal of Buddhist Art History] 14: 71–105.

Ji, Ziwei / Lee, Nayeon / Frieske, Rita / Yu, Tiezheng / Su, Dan / Xu, Yan / Ishii, Etsuko / Bang, Yejin / Madotto, Andrea / Fung, Pascale (2023): "Survey of hallucination in natural language generation", in: ACM Computing Surveys 55, 12: 1–38. DOI: 10.1145/3571730.

Jung, Song-yi (2023): "Meta archive in the fields of arts and culture" [예술문화 분야에서의 메타 아카이브], in: Journal of Museum Studies [박물관학보] 45: 239–263.

Muller, Charles (2009): "The Digital Dictionary of Buddhism [DDB]: Present Status and Future Developments", in: Japanese Studies Around the World 15: 87–100.

Nagasaki, Kiyonori (2015): "SAT大藏經テキストデータベース 人文学におけるオープンデータの活用に向けて", in: Journal of Information Processing and Management 58, 6: 422–437.

Natale, Simone / Surace, Bruno / Mensa, Enrico / Befera, Luca (2025): "ChatGPT for cultural heritage and the customization of generative AI: A talkthrough analysis of the Luigi Einaudi chatbot", in: New Media & Society: 1–26. DOI: 10.1177/14614448251384258.

Nazar, Rogelio / Acosta, Nicolas (2023): "Termout: A tool for the semi-automatic creation of term databases", in: Haddad Haddad, Bassam / Rigouts Terryn, Els / Mitkov, Ruslan / Rapp, Reinhard / Zweigenbaum, Pierre / Sharoff, Serge (eds.): Proceedings of the Workshop on Computational Terminology in NLP and Translation Studies (ConTeNTS). Varna: INCOMA Ltd 9–18.

Park, Eun-hwa (2013): "The feature of the gwaebul (掛佛) representing the crowned bodhisattva style (戴冠菩薩形) in Chungcheong Province in the 17th century of the Joseon Dynasty" [조선 17세기 충청권역 戴冠菩薩形 掛佛의 특색], in: Munmul Yeongu [문물연구] 23: 91–120.

Park, Jin-ho (2014): "Study on storytelling of digital museum 's exhibition content" [디지털박물관 전시콘텐츠 스토리텔링 연구: 전통문화콘텐츠박물관 전시콘텐츠 분석사례를 중심으로], in: Humanities Contents [인문콘텐츠] 33: 149–183.

Park, So-young (2012): "A study on hanging Buddhist pictures of Maitreya" [朝鮮後半期 彌勒佛掛佛圖 研究], in: Gangjwa Misulsa [강좌미술사, The Art History Journal] 39: 139–162.

Trichopoulos, George / Konstantakis, Markos / Caridakis, George / Katifori, Akrivi / Koukouli, Maria (2023): "Crafting a Museum Guide Using ChatGPT4", in: Big Data and Cognitive Computing 7, 3: 1–15.

Wang, Shanshan / Lu, Heng / Yao, Hong / Wu, Zhen / Hu, Chao / Xie, Xiaoyun / Yu, Xiaoyi (2025): “Cultural Heritage Assistant: A Lightweight Retrieval Augmented Generation Method Enhanced Vision-Language Model for Cultural Heritage”, in: Proceedings of the 2025 International Conference on Intelligent Computing (ICIC 2025): 1–14.

Yoo, Dong-hwan (2014): “A Study of Digital Museum based Digital Heritage”, in: The Korea Contents Society ICCS (2014): 67–68.

Yoo, Sang-woo / Kim, Se-ah (2023): “Reconceptualizing museum docents based on historical considerations”, in: Journal of Museum Studies 45: 141–175.

Standards & Ontologies

International Organization for Standardization (2023): Information and documentation — A reference ontology for the interchange of cultural heritage information (ISO 21127:2023; CIDOC CRM v7.1.3). <https://www.iso.org/standard/85100.html> [05.05.2026].

Miles, Alistair / Bechhofer, Sean (eds.) (2009): SKOS Simple Knowledge Organization System reference (W3C Recommendation, 18 August 2009). World Wide Web Consortium. <https://www.w3.org/TR/skos-reference/> [05.05.2026].

DCMI Usage Board (2020): DCMI Metadata Terms (Recommendation, 20 January 2020). Dublin Core Metadata Initiative. <https://www.dublincore.org/specifications/dublin-core/dcmi-terms/> [05.05.2026].

Brickley, Dan / Miller, Libby (2014): FOAF vocabulary specification 0.99 (Paddington edition, 14 January 2014). <http://xmlns.com/foaf/spec/> [05.05.2026].

Lebo, Timothy / Sahoo, Satya / McGuinness, Deborah (eds.) (2013): PROV-O: The PROV ontology (W3C Recommendation, 30 April 2013). World Wide Web Consortium. <https://www.w3.org/TR/prov-o/> [05.05.2026].

D'Arcus, Bruce / Giasson, Frédéric (2009): Bibliographic Ontology Specification (Revision 1.3, 4 November 2009). <http://bibliotek-o.org/bibo/> [05.05.2026].

Web Resources

Academy of Korean Studies (n.d.): Encyclopedia of Korean Culture [한국민족문화대백과사전]. <http://encykorea.aks.ac.kr/> [05.05.2026].

Academy of Korean Studies, Digital Humanities Lab (n.d.): CHAID ontology [Custom domain ontology]. <http://www.dh.aks.ac.kr/ontologies/CHAID#> [05.05.2026].

Digital Dictionary of Buddhism (n.d.): Digital Dictionary of Buddhism (DDB). <http://www.buddhism-dict.net/ddb/> [05.05.2026].

Google (n.d.): Google Arts & Culture. <https://artsandculture.google.com/> [05.05.2026].

Korea Heritage Service (n.d.): Korean Heritage Portal [국가유산포털].
<https://www.heritage.go.kr/> [05.05.2026].

Software

Chase, Harrison (2022): LangChain (Version 1.2.15) [Software].
<https://github.com/langchain-ai/langchain> [05.05.2026].

Google (2024): Google Generative AI Python SDK (Version 0.8.5) [Software].
<https://github.com/google/generative-ai-python> [05.05.2026].

Google (2025): Gemini 2.5 Flash [Large language model].
<https://deepmind.google/technologies/gemini/> [05.05.2026].

Neo4j, Inc. (2024): Neo4j (Version 5.26.0 LTS) [Computer software].
<https://neo4j.com/> [05.05.2026].

저자 소개

Song-yi Jung, a Ph.D. candidate, and Iro Lim, a master's student, are co-first authors conducting digital humanities research in Cultural Informatics at the Graduate School of Korean Studies, the Academy of Korean Studies, under the supervision of corresponding author Professor Baro Kim. Jung specializes in Korean art history and semantic data modeling, while Lim focuses on the design of human–AI collaborative systems for artistic and cultural domains, with both contributing to the study's humanistic and technological foundations.

Keyword

Cultural Heritage, AI Docent, Graph-RAG, Knowledge Graph, Digital Curation, Joseon Buddhist Painting

영문 초록 축약

This study presents a Graph-RAG-based AI docent prototype, validated through a Joseon-period hanging Buddhist painting. By grounding generative AI in a semantic knowledge graph and institutional guidelines, the system mitigates LLM hallucinations and overcomes physical signage limitations, demonstrating responsible curation for trustworthy, multimodal cultural heritage interpretation.

0503

999 words

Developing an AI Cultural Heritage Docent Web Prototype: Internalizing Institutional Guidelines for Responsible Curation

Background and Objectives

This study proposes a Graph-RAG-based AI Smart Docent web prototype that integrates a semantic knowledge graph with generative AI to overcome both LLM hallucination and the structural constraints of physical signage. Current cultural heritage information services face two structural limitations. First, traditional physical interpretive media remain confined to unilateral information delivery, providing uniform content despite the diversity of visitors' ages, prior knowledge, and languages. Second, general-purpose LLMs introduced as an alternative tend to generate factually inaccurate information due to their lack of domain-specific knowledge, thereby undermining the epistemic reliability required in cultural heritage interpretation.

Selection of the Empirical Subject

This study selects the Hanging Painting of Janggoksa Temple (Maitreya Buddha, 1673, National Treasure) as a single empirical case. The work depicts dozens of iconographic figures—buddhas, bodhisattvas, and arhats—on a single canvas, and its hwagi (畫記) explicitly records diverse donation categories (materials, pigments, ritual offerings, production stages) and a roster of participating monk-painters. This data composition, in which iconographic classification and person-relation information richly coexist, provides suitable conditions for validating the proposed integration of a thesaurus and a relation-expressive knowledge graph (Jeong, 2012; Park, 2012; Korea Heritage Service, 2022). The methodology is first validated through this work and then progressively extended to Buddhist sculpture, architecture, and archaeological sites, establishing a foundation for a generalizable generative docent model across Korean cultural heritage.

Methodological Orientation

This study formalizes humanities data—basic specifications, hwagi records, personal records, and thesauri—into RDF/OWL-based semantic data as a representational layer for scholarly knowledge sharing, and converts these data into a graph database for Graph-RAG inference. A generative AI Smart Docent model is then developed using verified data retrieved through natural language queries, and implemented as a web-based prototype.

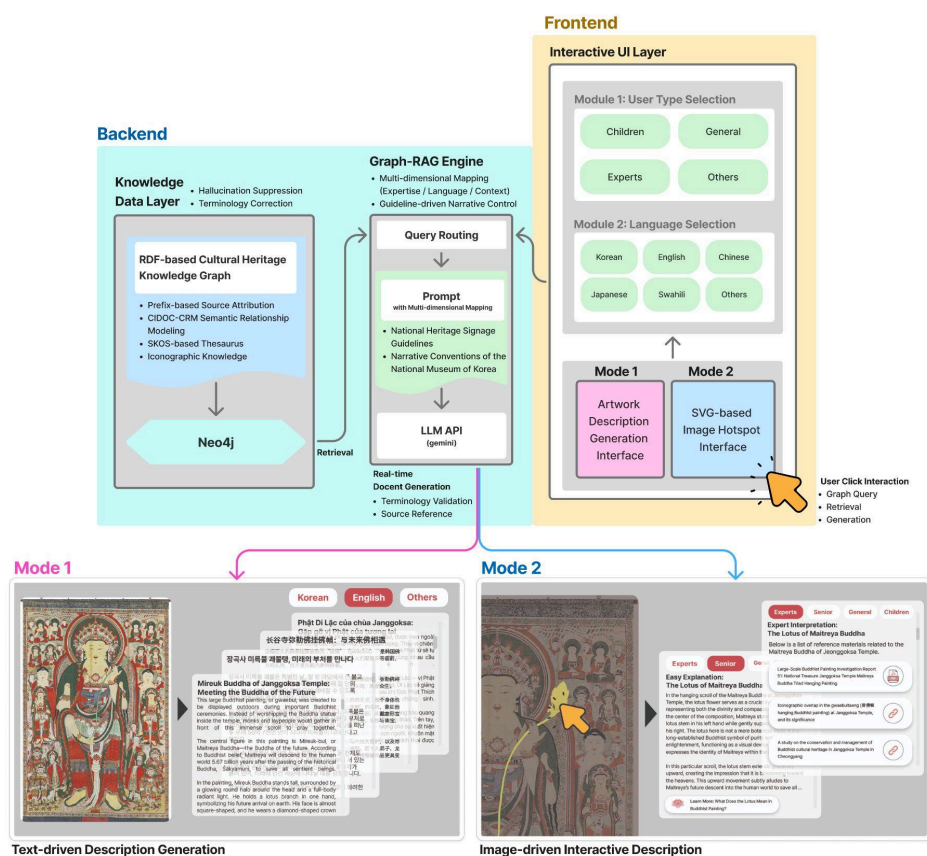
Related Work and Differentiation

This study builds on prior research that leverages digital technologies to expand museums beyond their physical boundaries, including Semantic Web-based meta-archives (Jung, 2023), immersive digital exhibitions in ubiquitous environments (Yoo, 2014), and personalized smart docent services (Gu & Shin, 2016). While generative AI has recently enabled highly personalized information delivery without pre-defined scenarios, its hallucination problem remains a critical limitation in domains requiring historical accuracy (Natale et al., 2025). Therefore, recent studies (MuseRAG++, Cultural Heritage Assistant) emphasize grounding responses in expert-verified data to ensure responsible curation and reliability (Trichopoulos et al., 2023; Nazar & Acosta, 2023; Wang et al., 2025; Hu, 2025).

Data Engineering and Knowledge Structuring

The core of this study lies in restructuring analog and unstructured source resources into machine-readable forms inferable by generative AI, thereby technically mitigating AI hallucination. (1) The signage guidelines issued by the Korea Heritage Service are

formalized as rule-based constraints embedded into the system prompt, specifying a priori the narrative principles the AI must observe (Korea Heritage Service, 2024). (2) Four categories of heterogeneous data—basic specifications, hwagi records, personal records, and thesauri—are formalized as RDF/OWL ontologies, leveraging SKOS for the iconographic thesaurus and CIDOC-CRM for semantic relations among persons, works, places, and events. The formalized data are then converted and loaded into a property graph in Neo4j for Graph-RAG inference, with iterative refinement: load results are validated via Cypher and the RDF/OWL source is revised on discrepancies. This establishes a dual structure: RDF/OWL functions as the representational standard for knowledge modeling and scholarly sharing, while the Neo4j property graph supports real-time inference and query response. (3) Multilingual designations and alternative names (imyeong, 異名) are collected and mapped from authoritative sources including the Digital Dictionary of Buddhism (Muller, 2009), the Encyclopedia of Korean Culture, the Korean Heritage Portal, and the hwagi of the Janggoksa Hanging Painting of Maitreya Buddha (Korea Heritage Service, 2022). Each value receives a source-identifying prefix to secure scholarly accuracy and provenance, technically supporting the epistemic reliability of AI-generated interpretation.



The Graph-RAG of this study is a RAG variant that uses structured query results from the graph database as grounding sources, aligning LLM responses with verified facts (see Figure 1). The backend server classifies natural language queries by type and selectively executes

pre-defined Cypher queries, reliably retrieving core humanistic elements such as participating artists, dates, iconography, and hwagi records. In the generation stage, the LangChain framework integrates retrieved graph data into the prompt, with Google Gemini as the generative model. The system prompt incorporates the Korea Heritage Service guidelines, and the system adjusts narrative style and difficulty according to user expertise (children, general audiences, domain experts) and language preference, forming a customized generation mechanism in which the same grounding data can be reconfigured for different contexts.

Interaction and User Experience

Implemented with a Next.js front-end and a FastAPI back-end, the prototype offers a button-based interface for question selection and structured, data-grounded interpretation. The study further proposes an AI Smart Docent function in which core iconographic elements serve as SVG-based clickable images; selecting an element triggers real-time retrieval of associated graph data and generates a customized interpretation. This multimodal approach—integrating visual, textual, and structured data—lays a foundation for future extension to AR and 3D-based exhibition environments responsive to visitor interaction.

Conclusion and Expected Contributions

This study recontextualizes professional docent interpretation as a visitor-driven interactive experience. By formalizing dispersed humanities knowledge into a dual structure of RDF/OWL-based representation and graph-database-based inference with integrated real-time access, the system simultaneously supports scholarly knowledge sharing and AI-driven inference. By using verified semantic data as a grounding source to mitigate AI hallucination, the study presents the technical feasibility of responsible curation in digital environments. Ultimately, the model contributes a generalizable framework extensible to diverse cultural heritage domains, advancing the practical use of digital heritage and scholarly methodologies. The system's effectiveness, however, requires further validation through user evaluation and quantitative testing; the constructed RDF/OWL ontology and implementation code will be released with a forthcoming journal article to support reproducibility and collaborative extension within the digital cultural heritage research community.

References

- Books

Kim, J. (2019). *Hanguk Bulgyo Misulsa* [한국불교미술사, History of Korean Buddhist Art]. Sechang Publishing Company.

Kim, W., & Ahn, H. (2003). *Hanguk Misulsa* [한국미술사, History of Korean Art]. Sigongsa.

Korea Heritage Service. (2019). *Guidelines for Writing English Cultural Heritage Signage* (Rev. ed.).

Korea Heritage Service. (2022). *Daehyeong bulhwa jeongmil josa bogoseo 51: Gukbo Janggoksa Mireukbul gwaebultaeng* [대형불화 정밀조사 보고서 51 _ 국보 장곡사

미륵불괘불탱, Research report of large Buddhist painting gwaebultaeng 51: Hanging Painting of Janggoksa Temple (Maitreya Buddha)]. Korea Heritage Service.

Korea Heritage Service. (2024). *Gukga yusan annaepan tonghap garideuraen* [국가유산 안내판 통합 가이드라인]. Korea Heritage Service.

Moon, M. (2021). *Hanguk Bulgyo Hoehwasa* [한국불교회화사, History of Korean Buddhist painting]. Youlhwadang.

- Articles & Proceedings

Bonetti, A., Salcedo-Puche, A., Vila-Francés, J., Benavent-Garcia, X., Fernández-Vargas, E., Magdalena-Benedito, R., & Soria-Olivas, E. (2025). An Agent-Based RAG Architecture for Intelligent Tourism Assistance: The Valencia Case Study. Preprints. <https://doi.org/10.20944/preprints202511.0576.v1>

Gu, J., & Shin, D. (2016). The Importance of Robot Personality in a Museum Context [박물관 맥락에서의 로봇성격 설정의 중요성 연구]. *Journal of the Korea Contents Association* [한국콘텐츠학회논문지], 16(3), 184–197. <https://doi.org/10.5392/JKCA.2016.16.03.184>

Hu, Y. (2025). MuseRAG++: A Deep Retrieval-Augmented Generation Framework for Semantic Interaction and Multi-Modal Reasoning in Virtual Museums. Research Square. <https://doi.org/10.21203/rs.3.rs-7281889/v1>

Jeong, M. (2012). Iconographic overlap in the gwaebultaeng (掛佛幀, hanging Buddhist painting) at Janggoksa Temple, and its significance [<長谷寺 靈山大會掛佛幀>에 보이는 도상의 중첩과 그 의미]. *Bulgyo Misulsahak* [불교미술사학, Journal of Buddhist Art History], 14, 71–105.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), Article 248, 1–38. <https://doi.org/10.1145/3571730>

Jung, S.-Y. (2023). Meta archive in the fields of arts and culture [예술문화 분야에서의 메타 아카이브]. *Journal of Museum Studies* [박물관학보], 45, 239–263.

Muller, A. C. (2009). The Digital Dictionary of Buddhism [DDB]: Present Status and Future Developments. *Japanese Studies Around the World*, 15, 87–100.

Nagasaki, K. (2015). SAT大蔵経テキストデータベース 人文学におけるオープンデータの活用に向けて, *Journal of Information Processing and Management*, 58(6), 422–437.

Natale, S., Surace, B., Mensa, E., & Befera, L. (2025). ChatGPT for cultural heritage and the customization of generative AI: A talkthrough analysis of the Luigi Einaudi chatbot. *New Media & Society*, 1–26. <https://doi.org/10.1177/14614448251384258>

Nazar, R., & Acosta, N. (2023). Termout: A tool for the semi-automatic creation of term databases. In A. Haddad Haddad, A. Rigouts Terryn, R. Mitkov, R. Rapp, P. Zweigenbaum, &

S. Sharoff (Eds.), *Proceedings of the Workshop on Computational Terminology in NLP and Translation Studies (ConTeNTS) Incorporating the 16th Workshop on Building and Using Comparable Corpora (BUCC)* (pp. 9–18). INCOMA Ltd.

Park, E. (2013). The feature of the gwaebul (掛佛) representing the crowned bodhisattva style (戴冠菩薩形) in Chungcheong Province in the 17th century of the Joseon Dynasty [조선 17세기 충청권역 戴冠菩薩形 掛佛의 특색]. *Munmul Yeongu* [문물연구], 23, 91–120.

Park, J. (2014). Study on storytelling of digital museum's exhibition content [디지털박물관 전시콘텐츠 스토리텔링 연구: 전통문화콘텐츠박물관 전시콘텐츠 분석 사례를 중심으로]. *Humanities Contents* [인문콘텐츠], (33), 149–183.

Park, S. Y. (2012). A study on hanging Buddhist pictures of Maitreya [朝鮮後半期 彌勒佛掛佛圖 研究]. *Gangjwa Misulsa* [강좌미술사, The Art History Journal], (39), 139–162.

Trichopoulos, G., Konstantakis, M., Caridakis, G., Katifori, A., & Koukouli, M. (2023). Crafting a Museum Guide Using ChatGPT4. *Big Data and Cognitive Computing*, 7(3), 1–15.

Wang, S., Lu, H., Yao, H., Wu, Z., Hu, C., Xie, X., & Yu, X. (2025). Cultural Heritage Assistant: A Lightweight Retrieval Augmented Generation Method Enhanced Vision-Language Model for Cultural Heritage. In *Proceedings of the 2025 International Conference on Intelligent Computing (ICIC 2025)* (pp. 1–14).

Yoo, D. (2014). A Study of Digital Museum based Digital Heritage. The Korea Contents Society ICCS (2014), 67–68.

Yoo, S., & Kim, S. (2023). Reconceptualizing museum docents based on historical considerations. *Journal of Museum Studies*, (45), 141–175.

- Standards & Ontologies

International Organization for Standardization. (2023). Information and documentation — A reference ontology for the interchange of cultural heritage information (ISO 21127:2023; CIDOC CRM v7.1.3). <https://www.iso.org/standard/85100.html> (Namespace prefix crm: <http://www.cidoc-crm.org/cidoc-crm/>)

Miles, A., & Bechhofer, S. (Eds.). (2009). SKOS Simple Knowledge Organization System reference (W3C Recommendation, 18 August 2009). World Wide Web Consortium. <https://www.w3.org/TR/skos-reference/> (Namespace prefix skos: <http://www.w3.org/2004/02/skos/core#>)

DCMI Usage Board. (2020). DCMI Metadata Terms (Recommendation, 20 January 2020). Dublin Core Metadata Initiative. <https://www.dublincore.org/specifications/dublin-core/dcmi-terms/> (Namespace prefix dcterms: <http://purl.org/dc/terms/>)

Brickley, D., & Miller, L. (2014). FOAF vocabulary specification 0.99 (Paddington edition, 14 January 2014). <http://xmlns.com/foaf/spec/> (Namespace prefix foaf: <http://xmlns.com/foaf/0.1/>)

Lebo, T., Sahoo, S., & McGuinness, D. (Eds.). (2013). PROV-O: The PROV ontology (W3C Recommendation, 30 April 2013). World Wide Web Consortium. <https://www.w3.org/TR/prov-o/> (Namespace prefix prov: <http://www.w3.org/ns/prov#>)

D'Arcus, B., & Giasson, F. (2009). Bibliographic Ontology Specification (Revision 1.3, 4 November 2009). <http://bibliotek-o.org/bibo/> (Namespace prefix bibo: <http://purl.org/ontology/bibo/>)

- Web Resources

Academy of Korean Studies. (n.d.). Encyclopedia of Korean Culture [한국민족문화대백과사전]. Retrieved from <http://encykorea.aks.ac.kr/> (Source-identifying prefix in this study: ekc)

Academy of Korean Studies, Digital Humanities Lab. (n.d.). CHAID ontology [Custom domain ontology]. Namespace: <http://www.dh.aks.ac.kr/ontologies/CHAID#> (Prefix chaid)

Digital Dictionary of Buddhism. (n.d.). Digital Dictionary of Buddhism (DDB). Retrieved from <http://www.buddhism-dict.net/ddb/> (Source-identifying prefix in this study: ddb)

Google. (n.d.). Google Arts & Culture. <https://artsandculture.google.com/>

Korea Heritage Service. (n.d.). Korean Heritage Portal [국가유산포털]. Retrieved from <https://www.heritage.go.kr/> (Source-identifying prefix in this study: khp)

- Software

Chase, H. (2022). LangChain (Version 1.2.15) [Software]. <https://github.com/langchain-ai/langchain>

Google. (2024). Google Generative AI Python SDK (Version 0.8.5) [Software]. <https://github.com/google/generative-ai-python>

Google. (2025). *Gemini 2.5 Flash* [Large language model]. <https://deepmind.google/technologies/gemini/>

Neo4j, Inc. (2024). *Neo4j* (Version 5.26.0 LTS) [Computer software]. <https://neo4j.com/>

Ramírez, S. (2024). FastAPI (Version 0.115.0) [Software]. <https://fastapi.tiangolo.com/>

Vercel. (2024). Next.js (Version 14.2.35) [Software]. <https://nextjs.org/>

Keyword

Cultural Heritage, AI Docent, Graph-RAG, Knowledge Graph, Digital Curation, Joseon Buddhist Painting

This study presents a Graph-RAG-based AI docent prototype, validated through a Joseon-period hanging Buddhist painting. By grounding generative AI in a semantic knowledge graph and institutional guidelines, the system mitigates LLM hallucinations and overcomes physical signage limitations, demonstrating responsible curation for trustworthy, multimodal cultural heritage interpretation.

연구 배경 및 목적

본 연구는 대규모 언어 모델(LLM)의 환각(hallucination) 현상과 기존 물리적 안내판의 구조적 제약을 극복하기 위해, 시맨틱 지식 그래프와 생성형 AI를 결합한 Graph-RAG 기반 AI 스마트 도슨트 웹 프로토타입을 제안한다. 현행 문화유산 정보 서비스는 두 가지 구조적 한계에 직면해 있다. 첫째, 기존 물리적 안내 매체는 일방향적 정보 전달에 머물러 관람객의 연령, 배경지식, 언어의 다양성에도 불구하고 동일한 내용만을 획일적으로 제공한다는 한계를 지닌다. 둘째, 그 대안으로 도입된 범용 LLM은 특정 도메인 지식의 부족으로 인해 사실과 다른 정보를 생성하는 환각 현상을 일으키며, 이는 엄격한 고증이 요구되는 문화유산 해설의 인식적 신뢰성(epistemic reliability)을 심각하게 저해한다.

실증 대상의 선정

본 연구는 국보 장곡사 미륵불 괘불탱(1673년)을 단일 실증 대상으로 선정한다. 본 작품은 단일 화면에 여래, 보살, 나한 등 수십 종의 도상 존상이 등장하고, 화기(畫記)에 재료, 안료, 의식 공양, 제작 공정 등 다양한 시주 항목과 참여 화승의 명단이 명확히 기록되어 있다. 이처럼 도상 분류 체계와 인물·관계 정보가 풍부하게 공존하는 데이터 구성은, 본 연구가 시도하는 시소러스와 관계 표현 지식 그래프의 연계 설계를 검증하기에 적합한 조건을 제공한다(정명희, 2012; 박수연, 2012; 국가유산청, 2022). 본 연구는 이 작품을 통해 방법론의 타당성을 일차적으로 검증하고, 이를 불교조각, 건조물, 유적 등으로 점진적으로 확장하여 한국 문화유산 전반에 적용 가능한 범용 생성형 도슨트 모델 구축의 토대를 마련하고자 한다.

연구의 방법론적 지향점

본 연구는 기본 제원, 화기(畫記) 기록, 인물 명세, 시소러스 등 이기종 인문 데이터를 RDF/OWL 기반 시맨틱 데이터로 정형화하여 학술적 지식 공유의 표상 계층을 마련하고, 이를 Graph-RAG 추론에 활용하기 위해 그래프 데이터베이스로 변환·적재한다. 나아가, 구축된 그래프 데이터베이스를 기반으로 Graph-RAG 기술을 활용하여 자연어 질의를 처리하고, 이를 통해 조회된 검증 데이터를 근거로 생성형 AI 스마트 도슨트 해설 모델을 개발한다. 또한, 해당 모델을 사용자가 직접 경험할 수 있도록 웹 기반 프로토타입으로 구현한다.

관련 선행 연구 및 차별성

본 연구는 디지털 기술을 매개로 박물관의 물리적 경계를 확장하고, 언어·연령·전문성이 상이한 관람객의 요구에 부응하는 사용자 맞춤형 지식 제공 모델의 기술적 토대를 구현하고자 다음과 같은 선행 연구의 성과를 참고한다. 대표적으로, 단순한 온라인 열람을 넘어 시맨틱 웹 개념을 바탕으로 분산된 데이터를 상호 연결하는 메타 아카이브 연구(Jung, 2023), 유비쿼터스 환경에서 시공간적 융합을 통해 몰입형 경험을 제공하는 디지털 전시 기획(Yoo, 2014), 그리고 스마트 도슨트를 위한 개인화 서비스 모델 구축(Gu & Shin, 2016) 등이 있다. 최근에는 생성형 AI의 등장으로 사전 정의된 시나리오 없이도 텍스트 생성이 가능해지면서, 고도로 개인화된 정보를 제공할 수 있는 새로운 가능성이 열렸다. 그러나 역사적 정확성이 필수적인 문화유산 분야에서 LLM의 환각 문제는 치명적인 한계로 지적된다(Natale et al., 2025). 이에 따라 MuseRAG++, Cultural Heritage Assistant 등 최근 연구는 도메인 전문가에 의해 검증된 고품질 데이터를 토대로 응답을 명시적으로 근거화함으로써, 책임 있는 큐레이션과 신뢰성을 확보하는 것이 필수적임을 강조하고 있다(Trichopoulos et al., 2023; Nazar & Acosta, 2023; Wang et al., 2025; Hu, 2025).

데이터 공학 및 지식 구조화

본 연구는 문화유산 도메인 지식과 AI 기술의 협업적 통합을 통해 수행된다. 그 핵심은 아날로그 및 비정형 형태의 원천 자료를 생성형 AI가 추론 가능한 기계 가독형(machine-readable) 형태로 재구조화하고, 이를 웹 환경의 RAG 모델로 구현하여 AI의 환각을 기술적으로 완화하는 데 있다.

문화유산 부문에서는 지식 공학적 설계를 통해 비정형 인문 지식을 기계 가독형 규범 및 시맨틱 데이터로 변환한다. (1) 국가유산청의 안내판 가이드라인을 규칙 기반 제약 조건으로 정형화하여 시스템 프롬프트에 반영함으로써, AI가 준수해야 할 해설의 서술 원칙과 논리적 위계를 사전에 명시한다(국가유산청, 2024). (2) 이어 기본 제원, 화기 기록, 인물 명세, 시소러스 등 4종의 이기종 데이터를 RDF/OWL 온톨로지로 정형화하며, 도상 시소러스에는 SKOS를, 인물·작품·장소·사건 간의 의미 관계 표현에는 CIDOC-CRM의 모델링 어휘를 활용하여 학술적 상호운용성을 확보한다. 정형화된 데이터는 Graph-RAG 추론을 위해 Neo4j 환경의 속성 그래프(Property Graph)로 변환·적재하며, 적재 결과의 검증과 RDF/OWL 원본 재정제를 반복하는 정제 절차를 거친다. 이를 통해 RDF/OWL은 지식 모델링과 학술적 공유의 기준 표상으로 기능하며, Neo4j 속성 그래프는 실시간 추론과 질의 응답을 담당하는 이중 구조를 구축한다. (3) 특히 지식 베이스 구축 과정에서 DDB(Digital Dictionary of Buddhism)(Muller, 2009), EKC(한국민족문화대백과사전), KHP(국가유산포털) 및 장곡사 미륵불 괘불탱 화기(국가유산청, 2022) 등 권위 있는 출처로부터 다국어 명칭과 이명(異名)을 수집·매핑한다. 이때 각 데이터 값에 원천 출처를 식별하는 접두어(prefix)를 부여하여 다국어 명칭의 학술적 정확성과 데이터 출처 추적성(provenance)을 확보함으로써, AI 해설의 인식적 신뢰성을 기술적으로 뒷받침한다.

시스템 구현 및 추론 파이프라인

본 연구는 설계된 시맨틱 지식 그래프와 생성 프롬프트를 결합하여 웹 기반 RAG 프로토타입을 구현한다. 본 연구의 Graph-RAG는 그래프 데이터베이스의 구조화된 질의 결과를 검색 증강의 근거로 활용하는 RAG 변형으로, LLM의 응답을 그래프상의 검증된 사실에 정렬(grounding)시키는 데 목적이 있다. 이를 위해 구축한 지식 그래프를 중심으로, 백엔드 서버가 자연어 질의를 입력받아 질문의 유형을 분류하고, 이에 따라 사전 정의된 Cypher 쿼리를 선택적으로 실행하는 라우팅 구조를 설계한다. 이러한 구조는 단순한 텍스트 검색이 아니라, 불화의 제작 참여 인물, 제작 시기, 도상 정보, 화기 기록 등 핵심 인문학적 요소를 안정적으로 식별하고 조회하기 위한 것이다.

본 시스템은 생성형 AI가 지식 그래프 데이터베이스에서 실제로 조회된 근거 데이터를 기반으로 해설을 구성하도록 설계된다. 이를 통해 생성 과정에서 발생할 수 있는 환각을 완화하고, 문화유산 해설에 요구되는 사실성과 신뢰성을 확보하고자 한다. 생성 단계에서는 LangChain 프레임워크를 활용하여 조회된 그래프 데이터를 생성 프롬프트에 통합하고, 생성 모델로는 Google Gemini를 사용한다. 시스템 프롬프트에는 국가유산청 가이드라인을 반영하여 AI가 준수할 해설 원칙을 명시한다. 또한, 사용자의 지식 수준(아동, 일반, 전문가 등)과 언어 조건에 따라 해설의 서술 방식과 난이도를 조절할 수 있도록 설계한다. 이러한 구조는 동일한 근거 데이터라 하더라도 수용자 맥락에 따라 상이한 방식으로 재구성·전달될 수 있도록 하는 맞춤형 생성 메커니즘을 형성한다.

인터랙션 및 사용자 경험

해당 시스템은 **Next.js** 기반 프론트엔드와 **FastAPI** 기반 백엔드로 구현된 웹 프로토타입 환경에서 버튼 기반 인터페이스로 구현되어, 사용자가 질문 유형을 직접 선택하고 이에 따라 구조화된 데이터 기반 해설을 제공받을 수 있도록 설계된다. 더 나아가, 불화의 도상 핵심 요소를 **SVG(Scalable Vector Graphics)** 기반의 클릭 가능한 이미지로 구성하여, 사용자가 특정 도상을 선택하면 해당 요소와 연관된 지식 그래프 데이터를 실시간으로 조회하고, 이를 토대로 맞춤형 해설이 생성되는 “**AI 스마트 도슨트**” 기능을 제안한다. 이러한 접근은 기존의 정적인 텍스트 중심 해설 방식에서 벗어나, 시각 정보, 텍스트 설명, 구조화된 데이터가 상호작용하는 멀티모달 해설 경험의 기초를 제시한다. 또한 본 연구에서 구현된 인터랙션 구조와 생성 모델은 향후 증강현실(**AR**) 및 **3D** 기반 전시 환경과 연계되어, 관람객의 행위에 따라 실시간으로 해설이 생성되는 확장형 문화유산 안내 시스템으로 발전할 수 있는 가능성을 지닌다.

결론 및 기대효과

결론적으로 본 연구는 일방향적 정보 전달에서 탈피하여, 도슨트의 전문적 해석을 관람객 주도의 인터랙티브 경험으로 재맥락화한다는 데 의의가 있다. 특히 물리적으로 분산된 이기종 인문 지식을 **RDF/OWL** 기반 표상과 그래프 데이터베이스 기반 추론의 이중 구조로 정형화하고 이에 대한 실시간 통합 접근을 가능케 함으로써, 학술적 지식 공유와 **AI** 기반 실시간 추론을 동시에 지원하며 관람객의 능동적인 지식 습득을 촉진한다. 나아가, 검증된 시맨틱 데이터를 그라운드링 소스로 활용하여 **AI**의 환각을 완화함으로써 디지털 환경에서도 “책임 있는 큐레이션”이 기술적으로 구현 가능함을 제시한다. 궁극적으로 본 모델은 향후 다양한 문화유산 도메인으로 유연하게 확장 가능한 범용적 기술 프레임워크를 제시함으로써, 디지털 문화유산의 실천적 활용과 학술적 방법론 확장에 기여할 것이다. 다만 본 연구에서 제안하는 시스템의 효용성은 향후 사용자 평가와 정량적 검증을 통해 입증될 필요가 있으며, 본 연구에서 구축한 **RDF/OWL** 온톨로지와 시스템 구현 코드는 후속 학술 논문 발표와 함께 공개하여 디지털 문화유산 연구 공동체의 재현성(**reproducibility**)과 협업적 확장에 기여하고자 한다.